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Review article

Indoor air quality in subway microenvironments: Pollutant characteristics, adverse health impacts, and population inequity

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ABSTRACT

Rapidly increasing urbanization in recent decades has elevated the subway as the primary public transportation mode in metropolitan areas. Indoor air quality (IAQ) inside subways is an important factor that influences the health of commuters and subway workers. This review discusses the subway IAQ in different cities worldwide by comparing the sources and abundance of particulate matter ($PM_{2.5}$ and PM_{10}) in these environments. Factors that affect PM concentration and chemical composition were found to be associated with the subway internal structure, train frequency, passenger volume, and geographical location. Special attention was paid to air pollutants, such as transition metals, volatile/semi-volatile organic compounds (VOCs and SVOCs), and bioaerosols, due to their potential roles in indoor chemistry and causing adverse health impacts. In addition, given that the IAQ of subway systems is a public health issue worldwide, we calculated the Gini coefficient of urban subway exposure via meta-analysis. A value of 0.56 showed a significant inequity among different cities. Developed regions with higher per capita income tend to have higher exposure. By reviewing the current advances and challenges in subway IAQ with a focus on indoor chemistry and health impacts, future research is proposed toward a sustainable urban transportation systems.

1. Introduction

Subway systems have emerged as the dominant public transport mode in many cities worldwide, and their number and size are increasing due to their convenience and cost-efficiency. By the end of 2020, subway systems served 193 cities worldwide, with a total length of over 17,000 km (International Union of Public Transport, 2022; Mohsen et al., 2018). The expansion of subways has also directly resulted in a substantial increase in the number of commuters over the past few decades (Gonzalez-Navarro and Turner, 2018; Yang et al., 2024). In Shanghai, the number of people traveling by subway now accounts for 70 % of the commuters of public transportation (SMTC, 2022).

Subways are characterized by a dense population of passengers, compact underground structures, and poor ventilation. This creates a unique indoor microenvironment for the buildup of emissions and the evolution of reactive airborne pollutants (Moreno and de Miguel, 2018; Passi et al., 2021a; Xu and Hao, 2017; Yang et al., 2023). Furthermore, passengers often spend a considerable amount of time in the subway, either in the station, on the platform, or on the trains. Although the total time of exposure may be short for each trip, frequent and repeated exposure to high concentrations of indoor pollutants can result in significant adverse health issues, especially for routine commuters (Frumkin, 2002; Ji et al., 2021b; Karlsson et al., 2005; Loxham and

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Nieuwenhuijsen, 2019; Martins et al., 2015; Nieuwenhuijsen et al., 2007; Roy et al., 2022). In addition, subway air pollutants may be an occupational health hazard for the staff, who are at higher risk due to longer periods of daily exposure, potentially for many years (Chillrud et al., 2005; Choi et al., 2019b; Ji et al., 2021b; Ji et al., 2022; Passi et al., 2021a). Due to the large population of passengers exposed to primary and secondary air pollutants in the subway microenvironment, indoor air quality (IAQ) of the subway system has become a topic of concern in socioeconomic sustainability, and public health.

Despite the high particulate matter (PM) concentrations and consequent exposure risks in subway microenvironments worldwide, a comprehensive understanding of the source, evolution chemistry, and toxicity of these airborne pollutants is still lacking (Gligorovski and Abbatt, 2018; Minguillon et al., 2018; Weschler, 2016). In addition, individual pollutant levels, the indoor chemistry among different airborne reactants, evidence of their health effects, and associated exposure inequity remain poorly understood. Thus, a review of subway IAQ is highly needed to provide a global overview of these research gaps. The present work is expected to provide the stakeholders, including city transit authorities, with information on state-of-the-art research findings. To achieve this goal, we have summarized the air pollutant concentrations and their main influencing factors inside subway systems, with special attention to underground platforms where most of the measurements were performed. The chemical composition of subway PM and the concentrations of gaseous pollutants, such as VOCs and SVOCs, have been reviewed regarding their roles in indoor chemistry and adverse health outcomes. The impact of regional socioeconomic factors, such as GDP, urban economic level, population density, geographical area, length of subway lines, and car ownership on accumulated personal exposure and health effects for passengers in different cities worldwide was also estimated. Finally, recommendations for future health-relevant studies and mitigation strategies on subway IAQ are proposed for a healthier and safer urban life.

2. Literature review methods

2.1. Article searching and screening

Data and information used in this review were gathered from research articles published from 2000 to 2024, which were identified through Web of Science, Google Scholar, Scopus, Springer, Elsevier, and Wiley Online Library searches. A total number of 1250 studies related to the subway microenvironment, indoor airborne pollution characteristics, influencing factors, and health effects were identified and collected using different keywords and their combinations (**Text S1**). 334 papers were excluded via the title screening procedure. After excluding 188 articles that are not relevant to subway IAQ based on full-text screening, we further screened 504 articles that are beyond the scope of the current review, such as articles with a focus on low reactivity pollutants or numerical modeling. As a result, a total number of 224 subway IAQ-related research papers were identified and analyzed in this review (Fig. 1).

2.2. Literature overview

As indicated in the world map (**Figure S1**), among all the countries/regions that conducted subway IAQ studies from 2000 to 2024, China published the most articles (29.3 %) on this topic, followed by South Korea and the United States with 17.7 % and 11.1 %, respectively. To further visualize this spatial information in a temporal manner, we summarized the number of papers related to subway IAQ published each year worldwide from 2000 to 2024 in each country/region (Fig. 2). From 2000 to 2024, the annual number of subway IAQ studies presented an increasing trend, with Asia, Europe, and North America showing the highest interest in this research field. For each article, we extracted and collected information including the study location, sampling method, and hazardous airborne pollutants in subway systems, explored their influencing factors, and evaluated their potential health effects on both commuters and subway workers.

3. PM in subway microenvironment

Due to the adverse health impacts on both commuters and subway workers, PM has been a key research topic among different subway IAQ studies (Passi et al., 2021a). The mean PM concentration measured at different sites around the world exhibits a significant variation between 1 to 1499 μg m^{-3} (**Table S1**), which can be affected by both indoor conditions (i.e., ventilation rate, piston effect, passenger numbers, rolling stock, wheels and brake pad composition) and outdoor air pollution (Hwang et al., 2016b; Kwon et al., 2008; Ma et al., 2014;

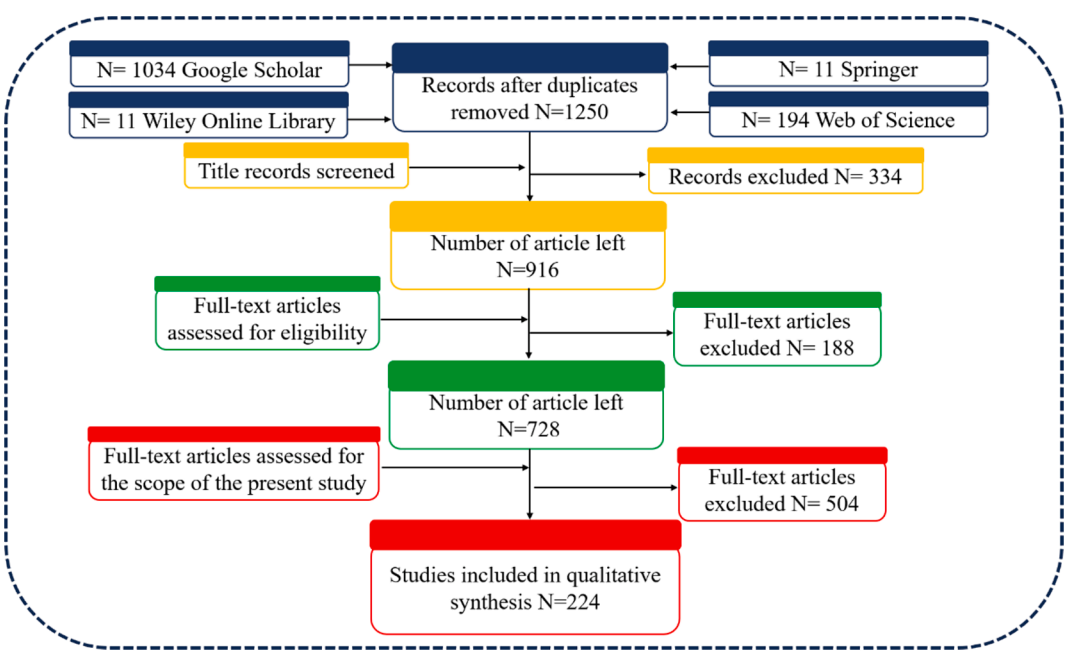

Fig. 1. Principles utilized in data selection and analysis in the present review.

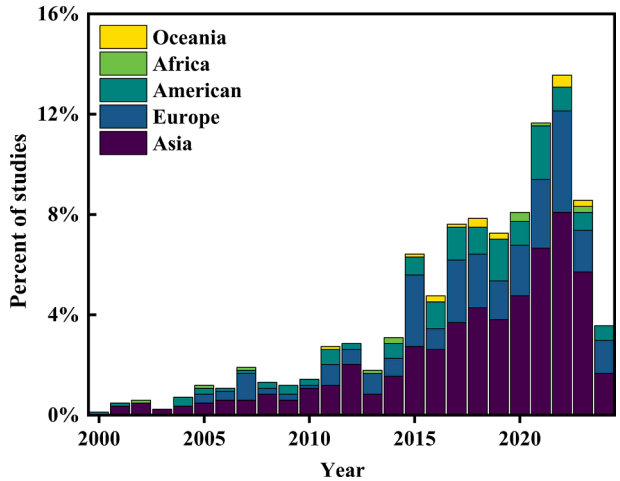

Fig. 2. The number of studies related to subway IAQ distributed in different cities around the world from 2000 to 2024.

Moreno et al., 2015; Reche et al., 2017). In **Table S1**, concentrations of ultrafine particles (UFP), submicron particles (PM_1), $PM_{2.5}$, and PM_{10} in subway systems of major cities around the world are summarized. Notably, the concentration of $PM_{2.5}$ is significantly higher compared to that of the outdoor environments (Ji et al., 2021b; Kwon et al., 2015; Smith et al., 2020; Yang et al., 2023). Especially during rush hours, the indoor-outdoor difference can even be more pronounced (Cheng et al., 2008; Kim et al., 2008; Li et al., 2007; Martins et al., 2015; Pan et al., 2019; Shakya et al., 2020; Wang et al., 2016c; Yang et al., 2023).

The physical design of the subway can influence PM pollution characteristics, including the depth of the subway, the type of tracks (Martins et al., 2015; Passi et al., 2021a), the design of the transfer stations (He et al., 2018; Passi et al., 2021a), and the presence platform screen door (PSD) systems (Kim et al., 2012; Kim et al., 2011; Moreno et al., 2017b). In addition, increasing wear between tracks, wheels, and brake pads resulting from aged infrastructure can cause further deterioration of IAQ (Van Ryswyk et al., 2017). In this section, the source and influencing factors of PM in the subway system are summarized to provide useful references for PM mitigation approaches.

3.1. Source and influencing factors of subway PM

Most subway stations have a semi-enclosed indoor microenvironment with passenger entrances and mechanical ventilation systems connecting to the outdoors. Thus, the sources of PM in subway stations are complex, but can be initially divided into two categories, indoor and outdoor, with indoor sources being the primary contributor for most cases (Huang et al., 2021; Huang et al., 2024; Juraeva et al., 2016; Martins et al., 2015). In addition to the pollutants carried by the airflow from the outdoor environment, indoor sources include wheel wear, rail wear, brake wear, resuspension, pantograph/catenary wear, and regular maintenance (Jung et al., 2010; Lee et al., 2018; Perrino et al., 2015; Sundh et al., 2009; Yang et al., 2023). The coarse component of PM_{10} has often been found to be the major contributor to subway PM, but there is also a non-negligible fraction of $PM_{2.5}$ (Kim et al., 2012; Liu and Chan,

Table 1
Influencing factors for subway platform PM and their specific impacts.

Influencing factor	Mechanisms	Impact on PM concentrations	Statistical significance*	References
Passenger numbers	Passenger body emission field (i.e. skin, hair, and clothes) Outdoor PM carried indoor by passenger flow	PM concentration increases during rush hour with large passenger flow (+)	Y	(Aksenov et al., 2023; Kwon et al., 2008; Ma et al., 2014)
Piston effect	PM resuspended in the air inside the tunnel due to incoming trains	PM concentration increases when trains are in operation (+)	N/A	(Martins et al., 2015; Moreno et al., 2014)
Depth of Platform	Deeper platforms have a lower air exchange rate with the outdoor environment	The deeper the platform (distant from tunnel entrnace or ventilation shaft), the higher the PM concentration (+)	Y	(Braniš 2006; Choi et al. 2019a; Ma et al., 2014; Saunders et al., 2019; Shakya et al., 2020)
System age	Resuspension of deposited dust in subway; system wear and tear	The system age has varied impacts on the subway PM concentration (+/-)	N/A	(Martins et al., 2016a; Van Ryswyk et al., 2021)
Outdoor environment	Outdoor PM level increase in pollution episodes	PM concentration in subway increases with outdoor PM levels (+)	Y	(Martins et al., 2015; Querol et al., 2012; Van Ryswyk et al., 2017; Zhang et al., 2022)
Maintenance and construction	Construction and maintenance using internal combustion engine machinery	PM increases with the emission from construction and maintenance machinery with internal combustion engines (+)	Y	(Choi et al., 2019b; Martins et al., 2015)
Train frequency	More wear and resuspension with higher frequency	PM concentration increases with a higher train frequency (+)	N/A	(Reche et al., 2017)
Power system	Wear of overhead wire or third rail	PM concentration increases (+)	N/A	(Martins et al., 2015; Querol et al., 2012)
Braking mechanisms	Mechanical brakes	PM concentration increases when mechanical brake applies (+)	N/A	(Qiao et al., 2015; Querol et al., 2012)
Train speed	Lower train speed with less friction for deceleration	PM concentration decreases (−)	N/A	(Martins et al., 2015; Shakya et al., 2020)
Number of line interchanges	Stations with the line interchanges have a larger indoor area and interconnected tunnels	PM concentration increases with the number of interchanges (+)	N/A	(Mak et al., 2024)
PSD system	PSD can effectively isolate air from platforms and tracks	PM concentration, especially PM_{10}, decreases with installing PSD (−)	Y	(Kim et al., 2011; Querol et al., 2012)
Heating Ventilation and Air Conditioning (HVAC) system	The operation of the air recirculation system influences the filtration rate inside the station	PM concentration decreases if there is an effective filtration system in operation (−)	N/A	(Martins et al., 2015; Nieuwenhuijsen et al., 2007)
Cleaning activities	High frequency of cleaning operation	PM concentration decreases (−)	N/A	(Martins et al., 2015)

*Note: N/A: Statistical significance test has not been implemented in referred studies; Y: A statistical test has been implemented, and the significance has been proved in referred studies.

2022; Martins et al., 2016a). Many studies found a higher ratio between $PM_{2.5}$ and PM_{10} during peak hours (Kwon et al., 2008; Ma et al., 2014; Wang et al., 2016b), indicating that the higher frequency of ridership/train arrival schedule and a higher volume of commuters during rush hour may contribute to a higher $PM_{2.5}$ concentration. With a higher train arrival frequency during rush hour, the presence of the power system (i.e., overhead wire or third rail) and the braking mechanisms (i.e., pneumatic or electric brakes) also influence the PM level in the subway microenvironment (Martins et al., 2016a). In Table 1, common factors that influence PM levels positively or negatively in the subway system are summarized. While the general trends in the impact of various factors on PM concentrations are summarized, it is worth noting that multiple factors may have an interactive effect on the IAQ of a single subway station, making it challenging to analyze and predict the PM concentrations based on a single factor. For instance, subway stations with high indoor and outdoor air exchange rates often have lower PM concentrations when compared to those with less effective ventilation systems in the same region. However, this relationship could be reversed during an elevated outdoor pollutant event such as haze or wildfire events. Depending on the specific station characteristics, certain factors may also have different impacts on PM concentrations. The influence of these factors is discussed in more detail in Sections 3.2 and 3.3.

3.2. Influencing factors for PM in the train

Many studies have investigated both indoor and outdoor factors influencing PM concentrations in the train, including the train ventilation rate, passenger number, and the opening frequency of train cabin doors and windows. In the meantime, the influence of each factor will also depend on the train air exchange rates and filtration efficiency, the occupants, and the pollutant concentration outside the train (Alves et al., 2017; Cheng and Li, 2010; Cheng et al., 2012; Mammi-Galani et al., 2017; Wang et al., 2016b). When the train is running, PM can be generated from mechanical frictions (e.g. between the train and the track, the brake system, and the wear of overhead wire etc.). Some of the generated PM is deposited in the tunnel, which could be resuspended due to the turbulent movement of air when a train passes (i.e. the piston effect), eventually leading to the spikes of PM concentration in the cabin upon the opening of the door. On the other hand, the movement of passengers upon getting on and off the cabin, and the individual emission/oxidation cloud can also significantly change the pollution characteristics in the subway microenvironment (Aksenov et al., 2023; Zannoni et al., 2022). In previous studies, the concentration of in-train $PM_{2.5}$ and PM_{10} was found to increase by 12–23 % and 20–24 % upon the opening of the train door (Li et al., 2018; Pan et al., 2019; Park and Ha, 2008; Wang et al., 2016c).

The HVAC system influences the PM concentrations inside the train; $PM_{2.5}$ can be reduced by 30 % when the air conditioning and recirculation system is turned on due to the effect of the particle filters (Alves et al., 2017). The influence of the ventilation system is also reflected in the seasonal variation of PM concentration in the subway. For instance, the subway PM concentration in summer is often lower than that in winter because of a more efficient HVAC system during summertime (Furuya et al., 2001; Martins et al., 2015). However, some studies (Martins et al., 2015; Qiao et al., 2015) showed that although the ventilation system can effectively remove larger particles (i.e., PM_{10}), the filtration efficiency of small particles, especially UFP, is relatively low. As particles with smaller sizes and larger surface-to-volume ratios can penetrate a deeper region of the respiratory system (Kelly and Fussell, 2012), filtration efficacy, therefore, deserves more attention in the future.

3.3. Influencing factors for PM at platform

PM concentrations on the platform and in the station are also important for the health of commuters and especially subway workers. They can be influenced by many factors, including the design of the subway platform (track layout, station depth, and station layout), emissions from train mechanical movements, operation management details, and the surrounding outdoor environments. Impacts from each factor were evaluated and discussed in detail as follows.

3.3.1. The design of the subway platform

The design of a subway platform can be generally categorized into three types: on-ground, underground, and elevated (Van Ryswyk et al., 2017). The on-ground and elevated platforms are directly connected to the open atmosphere, where the PM generated during the subway operation is diluted by the outdoor air (Martins et al., 2016a; Van Ryswyk et al., 2017). However, for semi-closed underground platforms, indoor emissions accumulate in the microenvironment because of the limited air exchange by the ventilation systems (Park and Ha, 2008). Studies have shown that the concentration of $PM_{2.5}$ in the underground platform was 20 % − 50 % higher than that of the ground platform (Aarnio et al., 2005; Cheng et al., 2012; Cheng and Yan, 2011; Chillrud et al., 2004). Moreover, PM concentration increases in underground stations with a greater depth and a lower air exchange rate (Braniš 2006; Figueroa-Lara et al., 2019; Smith et al., 2020).

The layout of a station, such as island platforms vs. side platforms, double-tracked vs. single-tracked, and transfer stations vs. non-transfer stations (**Figure S2**), could be another factor impacting PM concentration. From a mechanical perspective, the piston effect generated by train movements within tunnels of different structures could have different impacts on PM in the platform. When the train approaches the station, it squeezes the air in the tunnel and pushes it towards the platform, leading to drastically declined IAQ on the platform. In a study at the Barcelona Metro station (Martins et al., 2015), it was found that the double-tracked platform had a higher PM concentration than single-tracked platforms, and the concentration on the same platform was not evenly distributed due to the piston effect. For a double-tracked platform, when the train starts to brake upon entering the station, the PM concentration at the rear and in front of the station platform is similarly higher than that in the middle of the platform due to the joint effect from the rear and the front of trains on both tracks (Wang et al., 2016c). However, for a single-tracked platform, the PM concentration at the rear of the platform is normally greater than that at the middle of the platform, and the lowest level is often measured at the front of the platform (Park and Ha, 2008). The concentration of PM can also be influenced by whether it is a transfer station or not (Pan et al., 2019). Previous data shows that the concentration of PM in interchange stations was generally higher than that in stations without interchange as interchange stations generally have more trains passing through and thus more emissions (Kwon et al., 2015; Mak et al., 2024). PSD was originally designed as a safety door between the subway rail and the platform to separate the passengers from the train tracks. It has been proven that installing the PSD (**Figure S2**) can also help with reducing the platform PM concentration due to the reduced air exchange between the subway track and the platform (Cha et al., 2019; Han et al., 2012; Kim et al., 2012; Querol et al., 2012). It should be noted that while PSD could reduce the concentration of PM, no significant differences in the $PM_{2.5}/PM_{10}$ ratio have been observed between systems with and without the PSD (Kim et al., 2012), for which future mechanistic studies should be performed.

In addition, the age of the subway also influences PM concentration (Moreno et al., 2014; Pan et al., 2019; Smith et al., 2020). In general, newer stations have less amount of PM compared to older stations with aged tracks and less efficient HVAC systems (Moreno et al., 2014). In some cases, the internal PM concentration of new stations can equilibrate to be comparable with old stations within years and change correspondingly when replacing the existing train mechanical facilities in the subway. A study on subway IAQ in Toronto observed similar $PM_{2.5}$ concentrations in two newly opened stations with the neighboring platforms installed in the 1960s, suggesting a localized source and fresh

emissions of subway platform $PM_{2.5}$ (Van Ryswyk et al., 2021). Establishing real-time PM monitoring network could shed light on the effect of different subway internal facilities, rational maintenance strategy, and IAQ in subway microenvironments.

3.3.2. Operation management

Operation management of metro systems has significant impacts on subway IAQ, which includes train arrival/departure frequency (Huang et al., 2021; Ren et al., 2022), the type of ventilation and air cleaning systems (Qiao et al., 2015; Ren et al., 2022; Vilcassim et al., 2014), and metro cleaning maintenance (Figueroa-Lara et al., 2019; Querol et al., 2012). Studies found that the concentration of PM changes with the arrival frequency of the train (Martins et al., 2015; Moreno et al., 2014). As discussed earlier, such a phenomenon happens because of particles produced from the braking, friction between the train and the track, the resuspended dust raised by the trains, and the outdoor PM carried by the passenger flow (Aarnio et al., 2005; Ma et al., 2014; Martins et al., 2015; Minguillon et al., 2018; Ren et al., 2022). Previous work showed that the PM concentration can be reduced by 60 % and 52 % with the application of efficient ventilation and the magnetic filter, respectively (Juraeva et al., 2016). Other equipment such as electrostatic precipitation (Kim et al., 2017), compartment air purifier, dust collector installed at the bottom of the train (Sim et al., 2017), and vegetation biological filter (Kim et al., 2021) also exhibit effects in reducing indoor PM concentrations. Alternative operational measures, such as night-time maintenance and cleaning activities can also influence the PM levels in subway microenvironments. A previous study detected a rise in PM from 2:00 a.m. to 3:00 a.m., which resulted from the night maintenance of the local subway (Martins et al., 2015). Regular cleaning, such as mopping and vacuuming, can also effectively reduce PM concentration in the subway (Braniš 2006; Johansson and Johansson, 2003; Mugica-Álvarez et al., 2012; Salma et al., 2007). It is thus highly suggested that the decision on subway operation strategy should be made considering subway IAQ.

3.3.3. Impacts from outdoor environment

It has been widely observed that PM on the platform exhibits a correlation with the pollution level outside due to the indoor-outdoor air exchange (He et al., 2018; Morawska et al., 2001; Naddafi et al., 2016; Wang et al., 2022). Some studies found that the subway PM concentration during haze events was 2–3 times higher compared to that of clean days, indicating that the outdoor conditions could be another important factor affecting subway IAQ (Zhang et al., 2022). In addition, the PM concentration inside the subway is also related to the geographical location of the subway station. A previous study measured three subway platforms in Shanghai and found higher PM concentrations at two urban stations compared to the suburban one (Lu et al., 2015), which can be mainly attributed to the heavy traffic flow and high passenger volumes around the two urban stations. High concentrations of PM in the surrounding environment can enter the subway station through the ventilation system and passenger flow.

3.4. PM Indoor/Outdoor ratio

The indoor/outdoor ratio (I/O ratio) normally serves as an indicator for the source and indoor/outdoor air exchange efficiency of pollutants inside the subway. In general, due to the accumulation of indoor emissions and outdoor pollutants, the concentration of PM indoor is normally higher than that of outdoor (i.e., I/O > 1) (Carteni et al., 2015; Ji et al., 2021b; Shakya et al., 2020; Son et al., 2014). As summarized in **Table S2**, the I/O ratio of PM with different sizes (i.e., PM_1, $PM_{2.5}$, PM_{10}), particulate metals, and semi-volatile pollutants (i.e. aromatics) are often found to be larger than 1. For example, the I/O of $PM_{2.5}$ was measured to be 1.8–4.5 in Mexico City (Figueroa-Lara et al., 2019) and the I/O of PM_{10} was measured to be 2.1–3.5 in Seoul (Kwon et al., 2015). More recently, studies have measured the elemental composition of $PM_{2.5}$ inside/outside the subway in Nanjing, and systematically reported the I/O ratio (Ji et al., 2021b; Ji et al., 2021c). Even though not all the elements demonstrated an I/O ratio higher than 1 (i.e., P, S, Na, Ca, Cl), it should be noted that most of the redox-active transition metals, such as Fe, Cu, and Mn, exhibited a significantly higher level in subway indoor microenvironment compared to the outdoor environment (0.55–3070.06, 0.49–1176.36, and 0.54–331.54 for Fe, Cu, and Mn, respectively) (**Table S2**), highlighting the critical significance of regulating indoor source for these metallic components out of a public health perspective. There should be more attention paid to the I/O ratio given that it normally serves as an indicator of the relative importance of local sources compared to air exchange in the subway microenvironment, which can help the identification of effective IAQ mitigation policies.

4. Aerosol composition

In addition to the mass concentration of PM, other subway air pollutant metrics can be divided into four major categories, including metallic aerosol (Fe, Cu, Zn, and Mn etc.), non-metallic aerosol (black carbon, organic aerosol, SO_4^{2-}, and NO_3^- etc.), gaseous pollutants (VOCs), and bioaerosol (virus, bacteria, and fungi etc.). Exposure to these air pollutants on a daily basis may cause an increase in chronic and acute diseases, such as respiratory diseases (Lundstrom et al., 2011), cardiovascular outcomes (Chuang et al., 2020; Yang et al., 2018; Zhang et al., 2019), and cancer (Jung et al., 2012; Kam et al., 2013; Roy et al., 2020; Roy et al., 2022; Roy et al., 2019). However, the source, evolution pathway, and health effects of the toxic pollutants inside the subway microenvironment remain poorly understood at the present stage.

4.1. Metallic aerosol

Due to the sources of PM in subway microenvironments, such as the friction of train wheels and the brake pad, trace metals are accumulated and easily elevated in subway systems. Although these elemental compositions only constitute less than 50 % of total PM underground, they play a key role in source identification and health impacts of subway PM (Huang et al., 2024; Karlsson et al., 2005; Li et al., 2023a; Martins et al., 2016a; Roy et al., 2022). It should be noted that the train wheels, brake pads, overhead power lines, and rails all contain large percentages of metallic materials (Ding et al., 2009; Loxham et al., 2013), such that trace metals such as Fe, Cu, Mn, Ca, Al, Mg, Zn, Cr, Ni, Pb, and Hg have been frequently detected in PM underground (**Table S3**) (Aarnio et al., 2005; Azad et al., 2023; Cusack et al., 2015; Furuya et al., 2001; Kim et al., 2010; Loxham et al., 2013; Martins et al., 2016a; Minguillon et al., 2018; Park et al., 2019; Salma et al., 2007). The most widely applicable analytical methods for detecting elements in subway PM are Inductively Coupled Plasma Mass Spectrometry (ICP-MS) (Hwang et al., 2021; Zhang et al., 2020a), Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES) (Huang et al., 2021; Wang et al., 2016c), Scanning Electron Microscope (SEM) (Querol et al., 2012), and X-Ray Fluorescence (XRF) (Azad et al., 2023; Furuya et al., 2001).

Among all the elements detected in subways, Fe is the most abundant metallic component (Colombi et al., 2013; Lu et al., 2015; Martins et al., 2016b; Mohsen et al., 2018; Murruni et al., 2009; Qiao et al., 2015; Sheikh et al., 2022; Zhang et al., 2022). It was found that Fe mainly exists as iron oxides (α-Fe_2O_3, γ-Fe_2O_3, and Fe_3O_4) in subway PM, which is an important component of hematite and magnetite (Guo et al., 2014; Kang et al., 2008; Lu et al., 2015; Minguillon et al., 2018). There is no significant difference in the form of iron observed in particles in different fractions, indicating a relatively simple source of particulate iron (Moreno et al., 2015). Iron-rich particles in the subway mainly exist in two forms: one is uniform iron-rich fragments and the other is heterogeneous iron-rich fragments (Lu et al., 2015). The intense friction between the train and the track generates iron-rich particles via wear oxidization or rust, during which the generated iron-rich particles will later mix to form heterogeneous iron-rich fragments (Moreno et al., 2015).

As summarized in Fig. 3, the abundance of Fe and Cu in studies across the world span a wide range with 1.4–183.6 μg/m^3 and 0.007–3.0 μg/m^3, respectively (Colombi et al., 2013; Furuya et al., 2001; Huang et al., 2021; Ji et al., 2021a; Roy et al., 2022; Zhang et al., 2020a). Many other metallic elements were measured inside the subway microenvironment (**Table S3**). A recent study investigating the source of metallic PM elements in the Barcelona subway system identified steel wear as the main source of As, Hg, Sb, and Mo while brake pad wear was the primary source of Ba, Zn, Sb, As, Zn, Mo, Hg, and Pb (Font et al., 2019; Huang et al., 2024). Cr, Ni, and V are widely found in lubricating oils and released from the oil heating process due to frictions (Krachler et al., 2003; Martins et al., 2016a; Moreno et al., 2017b; Qiao et al., 2015). Thus, regulating the ubiquitous metallic pollution in the subway microenvironment remains challenging. It should be noted that transition metals such as Fe and Cu can initiate redox cycles and cause oxidative stress *in vitro* once inhaled (Bates et al., 2019; Kam et al., 2011). So far, effective measures have been piloted to reduce the metallic content in PM, such as replacing conventional friction materials with low-metal materials like graphite pantographs/catenaries and rubber wheels as well as concrete rollers (Font et al., 2019; Van Ryswyk et al., 2017) while more mitigation strategies (e.g. magnetic removal techniques) for lowering subway metallic pollution should be carried out in the future.

4.2. Non-metallic aerosol

4.2.1. Carbonaceous aerosol

Black carbon (BC) has been detected as a common composition in subway PM (Apte et al., 2011; Choi et al., 2019b; Kumar et al., 2023; Li et al., 2015; Vilcassim et al., 2014). So far, the reported BC levels in different subway systems range from 0.0005 to 52.5 μg/m^3 (Aarnio et al., 2005; Apte et al., 2011; Choi et al., 2019b; Luo et al., 2023; Van Ryswyk et al., 2017; Vilcassim et al., 2014). The major sources of BC in the subway system are the graphite link between the third rail and the train (Van Ryswyk et al., 2017) and the diesel maintenance machinery at night (Aarnio et al., 2005; Choi et al., 2019b; Vilcassim et al., 2014). Besides, outdoor environment could be another source of BC, such as traffic engine exhaust, biomass burning, and industrial coal combustion (Apte et al., 2011; Hankey and Marshall, 2015; Li et al., 2015; Shakya et al., 2020). Due to the formation mechanism, BC can act as a carrier of PAHs, triggering cardiopulmonary and neurological diseases (Chowdhury et al., 2022; Cui et al., 2020; Gan et al., 2011; Johansson et al., 2018; Lin et al., 2011; Ljungman et al., 2019; Shiraiwa et al., 2012). Airborne BC has been studied to affect oxidative stress, inflammatory cytokine secretion, and gene expression *in vitro*, which are associated with PM size, morphology, organic coatings, and atmospheric aging conditions (Hakkarainen et al., 2022; Li et al., 2013; Liu and Chan, 2022). A significant correlation (R^2 = 0.74) was observed between particulate BC and ROS in the Los Angeles Metro system (Kam et al., 2011). It should be noted that the lifetime of BC inside the semi-closed subway microenvironment could be even longer than that in the outdoor environment due to insufficient ventilation.

Organic aerosol (OA) is a significant content (20–90 %) in ambient particles (Jimenez et al., 2009), which has been recognized as an important component in subway PM in recent years (Figueroa-Lara et al., 2019; Huang et al., 2021; Luglio et al., 2021; Martins et al., 2016b; Van Drooge et al., 2018). The measured OA (normally measured as organic carbon, OC) ranged from 13.09 to 177 μg/m^3 in subways around the world (Figueroa-Lara et al., 2019; Huang et al., 2021; Luglio et al., 2021). Inhaling OA is associated with cardiorespiratory disease mortality across the United States (Pye et al., 2021). A significant correlation (R^2 = 0.92) was observed between OC and ROS in $PM_{2.5}$ collected at the Los Angeles Metro system (Kam et al., 2011). However, detailed chemical composition and sources of OA underground remain largely unresolved. Only compounds like particle-bound PAHs have been quantitively investigated on a molecular level (Kam et al., 2013; Martins et al., 2016b; Van Drooge et al., 2018). Cautions should be taken that substantial oxidants (i.e. O_3 and NO_x) and hydrocarbon precursors (i.e. VOCs) are present in the subway microenvironment (Feng et al., 2010), which indicates the potential for atmospheric secondary transformation. Subway is a very special indoor microenvironment with a relatively high pollutant mixing ratio, less efficient ventilation, low light intensity, and substantial human emission/oxidation field (Aksenov et al., 2023; Zannoni et al., 2022) such that the indoor chemistry could be very different from the outdoor environment. Future studies on physicochemical evolution and the corresponding health outcomes of subway carbonaceous aerosol with the advances in source apportionment methodologies are encouraged (Saraga et al., 2023).

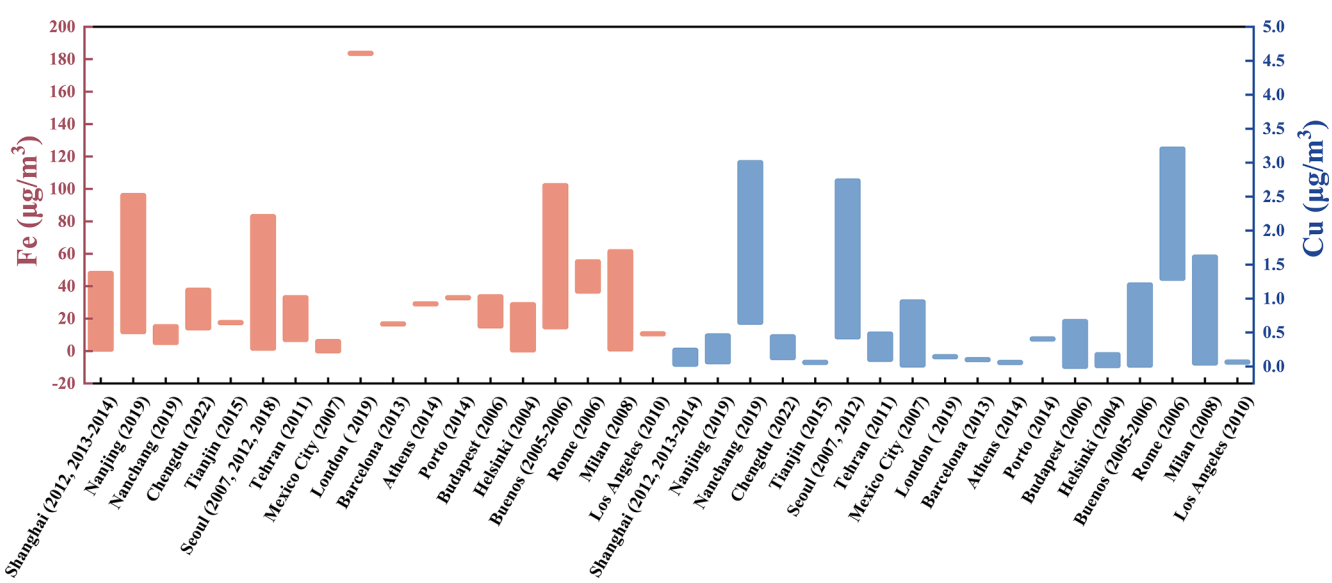

Fig. 3. Concentrations of Fe and Cu, two transition metals that have been most frequently detected in subway PM. Data were derived from previous work: Shanghai (Bao et al., 2014; Guo et al., 2014; Lu et al., 2015), Nanjing (Ji et al., 2021a; Ji et al., 2021b; Mao et al., 2019), Nanchang (Huang et al., 2021), Chengdu (Huang et al., 2024), Tianjin (Wang et al., 2016a), Seoul (Hwang et al., 2021; Kim et al., 2010; Park et al., 2014), Tehran (Kamani et al., 2014), Mexico City (Mugica-Álvarez et al., 2012), London (Smith et al., 2020), Barcelona, Athens, Porto (Martins et al., 2016a), Budapest (Salma et al., 2007), Prague (Cusack et al., 2015), Helsinki (Aarnio et al., 2005), Buenos (Murruni et al., 2009), Rome (Ripanucci et al., 2006), Milan (Colombi et al., 2013), and Los Angeles (Kam et al., 2011).

4.2.2. Inorganic secondary aerosol

The main inorganic compounds in subway PM are NO_3^- and SO_4^{2-}. By measuring the composition of subway $PM_{2.5}$, Huang et al. (2021) found that the concentration of NO_3^- was the highest (19.14 μg/m^3), followed by SO_4^{2-} (15 μg/m^3), which contributed 14 % and 11 % of total $PM_{2.5}$ mass, respectively. In Chengdu, the mass fraction of NO_3^- and SO_4^{2-} was found to be lower inside the subway (4 %, 4 %, 6 % for tunnel, carriage, and platform, respectively) compared to the outdoor environment (16 %) (Huang et al., 2024). The temporal distribution of nitrate and sulfate also exhibits seasonal variation. In Barcelona, SO_4^{2-} contributed to the highest underground $PM_{2.5}$ fraction (4–7 %) in warmer periods while NO_3^- was found to play a major role in colder periods (Martins et al., 2016b). NO_3^- and SO_4^{2-} can be directly transported from outdoor environment or formed secondarily from indoor SO_x and NO_x (Abadie et al., 2014; Fu, 2020; Huang et al., 2024). NO_x in the subway is mainly originated outdoors (Cao et al., 2017). The sulfur-containing species inside the subway system may come from the wear of sulphide-rich brake pads (Sb_2S_3 or MoS_2) or outdoor air (Font et al., 2019). Although sulfate and nitrate might not be toxic compounds (Park et al., 2018a), a recent laboratory work pointed out that those inorganic particulates could rapidly penetrate the surfactant layer and perturb the gene expression of the lung, affecting cell metabolisms such as tumorigenesis and immunity (Park et al., 2022).

4.3. Volatile organic compounds (VOCs)

In addition to the significant level of PM, VOCs exist as another IAQ issue in subway environments (Gong et al., 2017; Ji et al., 2021c; Lee et al., 2011; Li et al., 2006; Mao et al., 2019; Pang et al., 2020; Park et al., 2018b). VOCs in metro systems normally emerge from air exchange with the outdoor environment and emissions from indoor sources, such as furniture, building materials, inner surface paintings, human skin, and passenger activities (Aksenov et al., 2023; Gong et al., 2017; Ji et al., 2021c; Saraga et al., 2023). Fig. 4**a** summarizes the mean values for total VOCs (TVOCs) measured at different metro systems, which were found to be in the range of 84.4–1414.7 μg/m^3. The measured subway TVOCs exhibited seasonal variations trends with higher abundance in warm seasons and lower abundance in cold seasons (Chen et al., 2016). In the station offices in the Nanjing subway (Ji et al., 2021c), high levels of TVOCs were measured with an upper limit of nearly 3000 μg/m^3 during non-peak hours and 10000 μg/m^3 during rush hours, which was significantly higher than the national guideline (600 μg/m^3) and the outdoor VOCs background level (around 80–600 μg/m^3) in China (Zhang et al., 2017), indicating a significant indoor source of subway VOCs. A previous study observed that the concentrations of VOCs in older subways were 1–2 times higher than that in newer ones (Gong et al., 2017), which may result from passengers and the effectiveness of ventilation systems. The I/O ratios of different types of VOCs were often measured to be above 1 (**Table S2**). The I/O ratios for benzene, toluene, ethylbenzene, and xylenes (BTEX) at subway platforms in Shanghai were measured to be 1.4–2.1 during rush hour (Zhang et al., 2012). By collecting tube samples, Feng et al. (2010) reported an I/O ratio ranging from 0.6 to 3.4 for carbonyl compounds in Shanghai subways. The types of VOCs in subway microenvironments cover a wide range of volatilities, which can be further categorized into low-volatile organic compounds (LVOCs, $Log_{10}C^*<-0.5$), semi-volatile organic compounds (SVOCs, $-0.5 < Log_{10}C^*<2.5$), intermediate-volatile organic compounds (IVOCs, $2.5 < Log_{10}C^*<6.5$) and VOCs (Donahue et al., 2012). In Fig. 4**b**, the volatility distribution of the 10 volatile organics that have been frequently observed in subway systems is illustrated. However, the source, detailed composition, and transformation of those abundant volatile organics in typical subway microenvironments remain poorly understood. The dynamic gas-surface partitioning of VOC molecules with different volatilities/polarities (Wang et al., 2020), their reaction with indoor oxidants to form secondary organic aerosol (Aksenov et al., 2023; Zannoni et al., 2022), and the exposure risk of the secondary transformation products in subway microenvironment need further in-depth investigations (Rosales et al., 2022).

4.3.1. Carbonyls

Carbonyls are VOCs that have been less frequently identified in the subway (Feng et al., 2010; Pang and Mu, 2007; Xu and Hao, 2017). The concentration of carbonyl compounds is higher in rush hours, indicating that the carbonyl compounds in the subway are prone to be influenced by human activities (Aksenov et al., 2023). This is in agreement with previous observations showing that the mixing ratio of acetone and acrolein increased from 7.71 μg/m^3 to 26.28 μg/m^3 when the passenger number increased from 40 to 200 during rush hour (Gong et al., 2017). Carbonyls are known as an irritator and carcinogen (Feng et al., 2005). Electron-deficient carbonyls can interact with other organic functional groups and become active electron receivers while causing oxidative stress *in vitro* (Jiang et al., 2017). Unsaturated carbonyls are likely to react with cysteine residual and affect protein expression at a post-translational level (Han et al., 2020). Integrating pentafluorophenyl hydrazine (PFPH) coated solid sorbent and GC–MS for a better

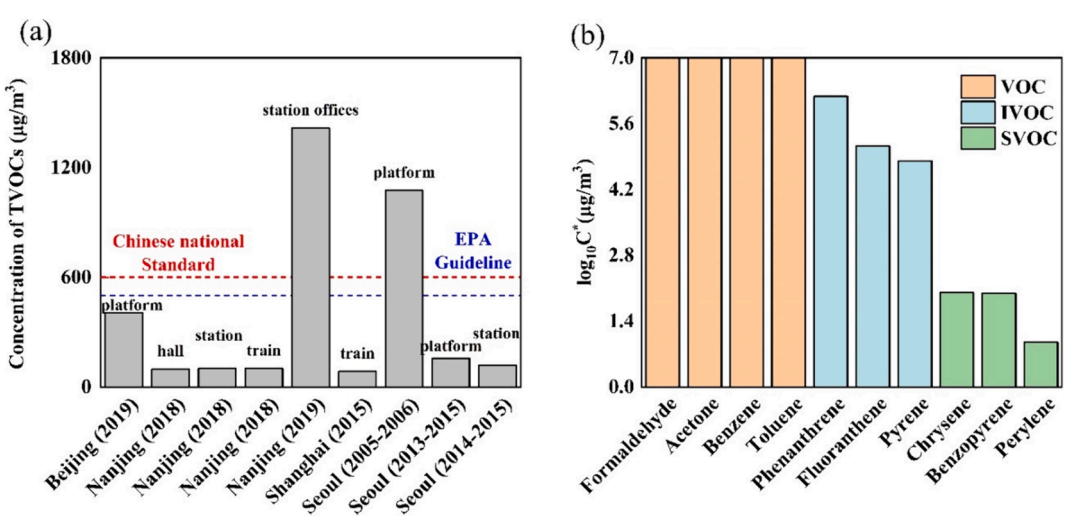

Fig. 4. (a) Level of TVOCs in subways around the world. (b) Volatility of typical volatile organic species in subway microenvironment. Data in (a) were derived from previous work: Beijing (2019) (Pang et al., 2020), Nanjing (2018) (Mao et al., 2019), Nanjing (2019) (Ji et al., 2021c), Shanghai (2015) (Gong et al., 2017), Seoul (2005–2006) (Lee et al., 2011), Seoul (2013–2015) (Park et al., 2018b), and Seoul (2014–2015) (Hwang and Park, 2019).

resolution and lower detection limit, Feng et al. (2010) found that the I/O ratios of most carbonyl compounds in the subway were greater than 1 (**Table S2**), such as formaldehyde, acetaldehyde, and acrolein. Future studies are encouraged to investigate the specific sources, indoor chemistry, and health impacts of carbonyl compounds in subways.

4.3.2. Polycyclic aromatic hydrocarbons (PAHs)

With two or more condensed ring structures, PAHs are a class of highly carcinogenic and mutagenic organic pollutants in the atmosphere (Dat and Chang, 2017; Lin et al., 2015; Mesquita et al., 2014). As summarized in **Table S4**, phenanthrene, pyrene, xylene, benzopyrene, and chrysene are PAHs that have been frequently detected in subway microenvironments (Bergvall and Westerholm, 2007; Chan et al., 2003; Martins et al., 2016b; Van Drooge et al., 2018). The abundance of PAHs in different subway systems spans a wide range (1–94 μg/m^3), which may result from indoor diesel machinery and air exchange. By investigating the origin of particulate PAHs in Barcelona subways using source apportionment, a previous study found that 75 % of the PAHs on platforms may have outdoor sources and 25% may have indoor sources (Van Drooge et al., 2018). It should be noted that most PAHs are SVOCs (Fig. 4**b**) that can readily undergo gas-particle partitioning in the subway microenvironment and participate in indoor chemistry. Enhanced oxidative stress and *in vitro* toxicity of aerosol evolved formed from PAHs after atmospheric aging have been revealed in many previous studies (Chowdhury et al., 2018; Han et al., 2020; Liu and Chan, 2022; Wang et al., 2018). As a consequence, it is critical to characterize the chemistry of PAHs and their functionalized derivatives underground from a public health perspective.

4.4. Bioaerosol

Bioaerosol refers to airborne virus, bacteria, fungi, and other microbial metabolism byproducts, which are ubiquitously detected in subway systems and often raise public health concerns (Dybwad et al., 2012; Grydaki et al., 2021; Metaxatos et al., 2021; Passi et al., 2021b; Ran et al., 2023; Triadó-Margarit et al., 2017). A previous study that investigated the public transit airborne microbiome found that bacteria is the most abundant (96.5 %), followed by virus (3.21 %), and fungi (0.19 %) (Leung et al., 2021). It is well known that the infection rates of virus are related to its airborne transmission (Wang et al., 2021). COVID-19 can remain active and spread in aerosol for several hours (Van Doremalen et al., 2020). It has been revealed that for every 1 μg/m^3 increase in $PM_{2.5}$ concentration, the mortality rate of COVID-19 will increase by 8 % (Wu et al., 2020). As a large underground network transportation system with significantly higher indoor PM concentration, the infection rate of respiratory virus in the subway microenvironment may be accelerated (Baron, 2020; Moreno et al., 2021; Wu et al., 2020). The main bacterial species that have been observed in the subway microenvironment are Bacillus, Staphylococcus epidermidis, and Micrococcus luteus (Dybwad et al., 2012; Gilleberg et al., 1998; Hernández-Castillo et al., 2014; Kim et al., 2011; Naddafi et al., 2011). In terms of fungi, Aspergillus, Penicillium, and Streptomyces are the main species (Gilleberg et al., 1998; Hernández-Castillo et al., 2014). Fig. 5 shows the abundance of fungi and bacteria in subway varies greatly across different cities. The subway bioaerosol level in some places are lower than the standards of American Conference Of Governmental Industrial Hygienists (ACGIH) and WHO (500 CFU/m^3), such as London (Gilleberg et al., 1998), Norway (Dybwad et al., 2012), Tokyo (Kawasaki et al., 2010), Istanbul (Metiner et al., 2021), and South Korea (Hwang and Park, 2019; Kim et al., 2022). On the other hand, the abundances of bacteria and fungi in subway stations in St. Petersburg (Bogomolova and Kirtsideli, 2009), Beijing (Dong and Yao, 2010), Korea (Cho et al., 2006), and Chennai (Passi et al., 2023) were found to exceed the guideline value from WHO and ACGIH (500 CFU/m^3).

Long-term exposure to bioaerosol has adverse effects on human health, which can result in a variety of diseases as well as immune system disorders. A previous study (Passi et al., 2021a) discovered that high passenger density is the main reason for the high bacterial concentration spread in the subway microenvironment. A few studies believe that the subway filtration system may be the main source of

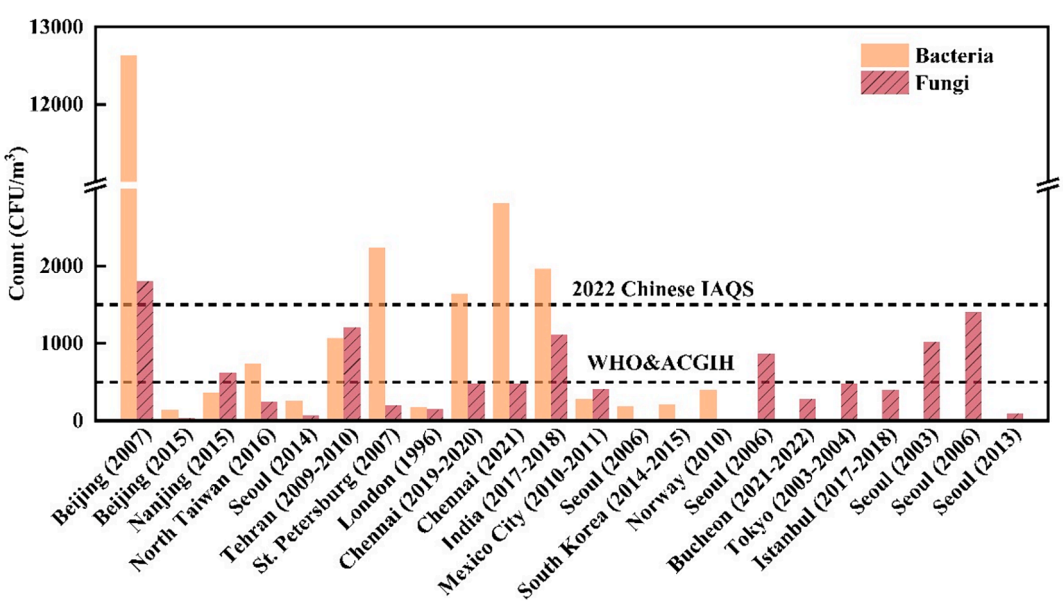

Fig. 5. Abundance of bacteria and fungi in different subway systems. Dahed lines are 500 CFU/m^3 and 1500 CFU/m^3 for bacteria and fungi given by WHO, American Conference of Govern-mental Industrial Hygienists (ACGIH), and the 2022 China Indoor Air Quality Standard (IAQS), respectively. Data were derived from previous work: Beijing (2007) (Dong and Yao, 2010), Beijing (2015) (Fan et al., 2017), Nanjing (2015) (Zhang et al., 2020b), North Taiwan (2016) (Chen et al., 2016), Seoul (2014) (Heo and Lee, 2016), Tehran (2009–2010) (Naddafi et al., 2011), St. Petersburg (2007) (Bogomolova and Kirtsideli, 2009), London (1996) (Gilleberg et al., 1998), Chennai (2019–2020) (Passi et al., 2023), Chennai (2021) (Passi et al., 2021b), India (2017–2018) (Madhwal et al., 2020), Mexico City (2010–2011) (Hernández-Castillo et al., 2014), Seoul (2006) (Hwang and Cho, 2016), South Korea (2014–2015) (Hwang and Park, 2019), Norway (2010) (Dybwad et al., 2012), Seoul (2006) (Hwang and Cho, 2016), Bucheon (2021–2022) (Kim et al., 2022), Tokyo (2003–2004) (Kawasaki et al., 2010), Istanbul (2017–2018) (Metiner et al., 2021), Seoul (2003) (Cho et al., 2006), Seoul (2006) (Hwang et al., 2016a), and Seoul (2013) (Hwang et al., 2016a).

fungi (Brincat et al., 2016; Cho et al., 2006; Passi et al., 2021a; Zhang et al., 2020b). Investigations in Mexico (Hernández-Castillo et al., 2014), London (Gilleberg et al., 1998), and Bangkok (Siriarchawatana et al., 2023) observed a good correlation between the abundance of bioaerosol inside the subway and indoor temperature/humidity. Human activity and geographical specificity may also affect the formation and fate of the microbial community (Kim et al., 2023; Leung et al., 2021). On the other hand, mechanical ventilation inside the subway has been proven to effectively remove microbial pollutants in the air (Hwang et al., 2016b). Zhang et al. (2020) found that after one hour of ventilation in the subway tunnel, the suspended bacteria and fungi decreased by 86.0 % and 28.7 %, respectively. Therefore, an efficient ventilation system with periodic cleaning and maintenance of filters in the subway is important for reducing the abundance of bioaerosol, at least for bacteria and fungi. On that basis, the source, fate, and adverse health effects of bioaerosol in subways need to be systematically evaluated for establishing future public health mitigation strategies.

5. Health effects of subway exposure

The essential purpose of studying subway IAQ is to protect commuters and subway workers from acute or chronic adverse health effects upon exposure underground (Canu et al., 2024). So far, using different semi-empirical risk estimation methods (**Table S5**), studies have illustrated the health risk of exposure to subway air pollutants based on the measured chemical species (e.g., metallic elements and PAHs) (He et al., 2018; Ji et al., 2022; Kam et al., 2013; Li et al., 2023b; Roy et al., 2022; Wang et al., 2016a; Yan et al., 2015). By performing chemical characterization and source apportionment of $PM_{2.5}$ in Nanjing, a recent work found that subway sources accounted for more than 50 % of carcinogenic and non-carcinogenic risks compared to other outdoor sources (Ji et al., 2022). So far, the main approaches for studying health outcomes of airborne pollutants in subway microenvironment can be categorized into laboratory and epidemiological investigations, which are reviewed as follows.

5.1. Laboratory observation

Oxidative potential (OP) indicates the capacity for pollutants to generate ROS and cause redox imbalance once inhaled (Bates et al., 2019). A recent study exposed 31 volunteers to subway air pollution and found that the measured acellular OP of PM at the stations exhibited significant associations with markers of the airway and nasal inflammation (Janssen et al., 2015). Notably, the OP of subway PM was found to be significantly higher compared to PM collected from other sources such as farms and traffic among different studies (Janssen et al., 2014; Jung et al., 2012; Kam et al., 2011; Moreno et al., 2017a; Steenhof et al., 2011; Strak et al., 2012). We collected OP results from three cell-free methods, namely, dithiothreitol (OP^{DTT}), electron spin resonance (OP^{EPR}), and ascorbic acid assay (OP^{AA}), for PM collected in different areas relative to the urban background level (Fig. 6). The observed OP values for PM in the subway system was significantly higher across all assays compared to PM from other sources (Fang et al., 2016; Janssen et al., 2014; Liu et al., 2014; Pietrogrande et al., 2018). Similarly, for cellular OP measured using the ROS-sensitive fluorescent probe (DCFH-DA), subway PM also exhibited significant oxidative stress (Jung et al., 2012; Kam et al., 2011). A recent study found that the OP of PM collected at the station with PSD was the lowest among all the studied stations, indicating that transition metals from the wear of train wheels and brakes can be the major contributor to the high OP of subway PM (Moreno et al., 2017a).

The release of reactive oxygen species (ROS) and significant oxidative stress caused by subway PM can be further confirmed by the stimulated expression of lactate dehydrogenase (LDH) *in vitro*, demonstrating its adverse impacts on the integrity of tissue and cell membranes (Seaton et al., 2005). Also, bioaerosol-induced ROS production varies with microbial type, species, and concentration (Samake et al., 2017). Another evidence for the *in vitro* toxicity of subway PM is the elevated expression of pro-inflammatory markers (Bachoual et al., 2007; Karlsson et al., 2006; Saunders et al., 2019; Seaton et al., 2005; Steenhof et al., 2011). Accompanied by the significantly reduced cell viability in the MTT assay, higher expressions of Tumor Necrosis Factor-alpha (TNF-α) and Macrophage Inflammatory Protein-2 (MIP-2) were observed with RAW264.7 cells exposed to coarse particle (2.5–10 μm),

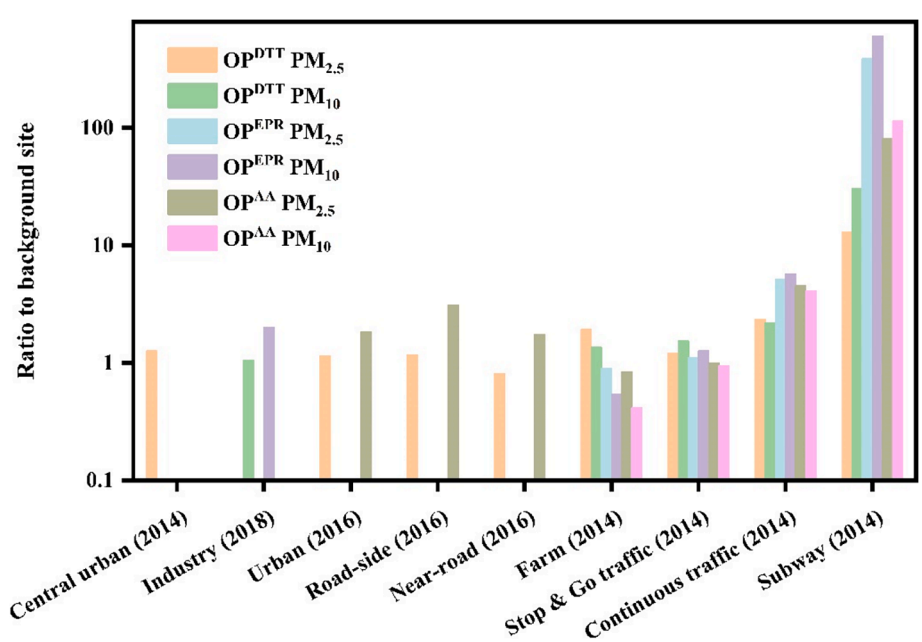

Fig. 6. Relative OP values measured using different OP assays (OP^{DTT}, OP^{EPR}, and OP^{AA}) for $PM_{2.5}$ and PM_{10} collected from farm, stop and go traffic, coutinuous traffic, subway station (Janssen et al., 2014), central urban (Liu et al., 2014), industry (Pietrogrande et al., 2018), urban, road-side and near-road (Fang et al., 2016) compared to respective background sites. PM collected in subway exhibited the highest OP than PM from other sources across different OP assays.

fine particle (<2.5 μm), and quasi ultrafine particle (qUF < 0.18 μm) collected at a subway station in Netherlands (Steenhof et al., 2011). By performing exposure to particles collected from the Paris subway, a former study reported an increased expression of matrix metalloprotease-12 (MMP-12) and heme oxygenase-1 (HO-1) (Bachoual et al., 2007). Integrating *in vivo* exposure and single-cell transcriptomics, a recent study highlights the roles of Toll-like receptors (TLRs) in subway $PM_{2.5}$-induced lung inflammation (Zhou et al., 2024). Recently, researchers found that subway $PM_{2.5}$ caused extrapulmonary organ injury related to the TLR/MyD88/NF-κB pathway (Wang et al., 2023). The genotoxicity of subway PM is another concerning aspect (Jung et al., 2012; Karlsson et al., 2008). By testing PM collected from different sources (wood combustion, a road simulator, street, and subway), Karlsson et al., (2005,2006) found that PM from the subway caused more DNA damage than the other particles ($p < 0.001$). In addition, evidence has been provided *in vivo* on the adverse health effects of subway PM, such as the enhanced level of cytokines and abnormal count of immune cells in bronchoalveolar lavage (Bachoual et al., 2007). In recent work, enhanced expression of platelet-activating factor receptor (PAFR) was found to be the major mechanism for pneumococcal infection upon exposure mice to PM collected in London (Miyashita et al., 2022).

Significant trace metal contents with redox-cycling capacities (i.e., Fe, Cu, Zn) were found in subway PM, especially Fe, which is responsible for its high toxicity (Chillrud et al., 2004; Kam et al., 2011; Karlsson et al., 2005; Sheikh et al., 2022; Strak et al., 2012). Meanwhile, the abundant content of PAHs (**Table S4**) can also contribute to the toxicity of subway PM (Jung et al., 2012). Oxygenated PAHs and nitrated PAHs formed during atmospheric aging processes may have a higher capacity to release ROS and induce other toxic effects (Han et al., 2020; Lammel et al., 2020; Wang et al., 2018). In addition, some studies have demonstrated that transition metals (i.e. iron) and organics in PM may have a synergistic effect on the toxicity, which should be further investigated for future mitigation strategies towards subway PM health effects (Xiong et al., 2017; Yu et al., 2018).

5.2. Epidemiological evidence

Epidemiological studies often indicate the associations between exposure to air pollutants and adverse health outcomes (Brauer et al., 2016; Cohen et al., 2017; Loxham and Nieuwenhuijsen, 2019). However, due to the lack of epidemiological linkages connecting specific diseases with exposure to air pollutants in subway systems, we are not fully aware of the health risks of exposure to pollutants in this unique microenvironment (Vilcassim and Thurston, 2023). So far, there is no consensus reached by epidemiological studies on the adverse health impacts of subway pollution. Several epidemiological studies reported insignificant associations between exposure to subway air pollutants and adverse health effects, such as lung function, relevant biomarkers, risk of lung cancer, and myocardial infarction. On the other hand, a few epidemiological studies showed evidence for the adverse health outcomes resulting from exposure to subway microenvironments. A randomized crossover study investigated the respiratory response of asthmatics and healthy individuals to subway air in Stockholm, and found that short-term exposure to subway PM may interfere with autonomic function in commuters while increasing the risk of cardiovascular disease. More recently, a study based on a 15,000-subway worker cohort in Parisian indicated that exposure to ambient PM_{10} may increase the risk of chronic obstructive pulmonary disease (COPD) (Canu et al., 2024).

Being specific about PM exposure from subway instead of outdoor air is a key feature of these epidemiological investigations. Using mixed-effect linear regression models and a microenvironmental exposure model, Van Ryswyk et al. (2017) estimated the contribution of subway $PM_{2.5}$ to daily personal $PM_{2.5}$ exposure in three Canadian cities (Toronto, Montreal, and Vancouver). In addition, performing epidemiological studies for subway exposure using personal sampling and measurements is also recommended for future studies as this route can provide detailed exposure information with high spatial and temporal resolutions (Bendl et al., 2023). As mentioned above, PM in subway stations also includes bioaerosol (Fig. 5), which can be sensitizing, toxic, or infectious, such as SARS-CoV-2 (Guzman, 2021). Studies have shown that about 30 % of respiratory diseases are caused by inhalation of aerosol (Chen et al., 2021). As a special indoor environment with semi-closed space and less efficient ventilation, public health challenges in the transmission of viable pathogens in the subway microenvironment have not been comprehensively characterized in existing literature. Given the large population of subway passengers, especially those living in cities with aged subway systems and heavy passenger flows, epidemiological studies connecting specific diseases and resolved pollutant characteristics (i.e., using source apportionment) inside subways are urgent.

5.3. Passenger exposure to subway PM

5.3.1. Exposure estimation and its influencing socioeconomic factors

In order to further explore the exposure of the global population to $PM_{2.5}$ in the subway, a quantitative analysis was conducted to calculate the accumulated exposure of subway passengers worldwide the spatial distribution and economic development, 13 cities, including Shanghai, Beijing, Barcelona, and Seoul were selected. Here, we included all studies with PM data during the recent ten years, where there are multiple studies on PM concentration (e.g., Shanghai conducted studies in 2013, 2018, and 2020), we used the latest data. Data on $PM_{2.5}$ concentration, population, and subway operation is collected for each city. The analysis focuses on city-wide subway passengers and only the average duration and exposure of passengers' stay on the platform was considered here as most of the $PM_{2.5}$ concentration data was monitored on platforms. With further data collection on the daily subway traveling characteristics, this study can be extended to cover the full trip underground, which includes both platform waiting and in-cabin traveling. The city-wide passenger exposure (Y) to the subway $PM_{2.5}$ on subway platforms during waiting time is estimated using Equation (1).

$$Y = C \times PS \times IR \times ET \tag{1}$$

Where C is the $PM_{2.5}$ concentration inside the subway of the corresponding city, $\mu g/m^3$, which was retrieved through a meta analysis (**Table S1**); *PS* is the number of subway travelers as a proportion of the total number of people traveling by public transportation in the city, in fraction, and the data comes from the transportation bureau operation reports of each city; *IR* is the respiratory rate per unit time, since subway commuting is defined as "light activity", the *IR* is 0.498 m^3/h according to the China Population Exposure Manual (Duan et al., 2015); *ET* is the exposure time in the subway. In this study, the exposure time is defined as the waiting time of passengers on the platform, which can be approximated as half of the arrival time interval, and it can be obtained from the local operator.

In order to illustrate the relationship between city-wide socioeconomic factors (details of data source can be found in Table S6) and PM concentration inside the subway system, we summarized the GDP, length of subway lines, urban population density, per-capita income and car ownership of these cities as independent variables, then performed spearman correlation coefficient analysis with subway $PM_{2.5}$. Table 2 shows the correlation and the level of significance between exposure and each independent variable.

Spearman correlation analysis shows that GDP, urban area, subway length, car ownership, per capita income, and exposure concentration are positively correlated. Among them, the correlations between exposure concentration with GDP and per-capita income are significant, respectively. For some developed cities, subways were built early, however, insufficient funds were invested in the later stages to update

Table 2
Spearman correlation coefficient of $PM_{2.5}$ exposure per-capita and different socioeconomic variables (*, $P < 0.05$).

Variable	Coefficient	*P* value
GDP (USD)	0.581	0.011*
Urban area (km^2)	0.170	0.499
Subway length (km)	0.414	0.088
Population density (people/km^2)	−0.051	0.842
Car ownership (10-thousand vehicles)	0.312	0.208
Per-capita income (USD)	0.519	0.027*

and maintain the equipment. The lack of maintenance towards the aging of internal ventilation systems and other facilities in subways, such as PSD, would lead to an increased exposure level for subway workers and commuters.

5.3.2. *Passenger exposure equity in the world*

The exposure inequity is depicted through a Lorenz curve and the Gini coefficient (Fig. 7). The cumulative $PM_{2.5}$ exposure is calculated with Eq. (1). By ranking cities in ascending order of per capita income, a Lorenz curve is plotted as the cumulative percentage of population versus the percentage of exposure. The exposure level via inhalation was also estimated based on the $PM_{2.5}$ concentration reported in each city and the semi-empirical exposure risk model Equation (2).

$$ECinh = C \times ET \times \frac{EF \times ED}{AT} \tag{2}$$

Where ECinh is exposure dose through inhalation (μg/m^3); ET is exposure duration on the platform (0.5 h/day); EF is exposure frequency of riders (252 days/year); ED is exposure duration of a rider in years; AT is averaging time (hour) (Roy et al., 2022).

As depicted in Fig. 7, among the studied cities, the population has a great disparity in PM exposure inside the subway. Based on the Barcelona point, about 69.4 % of people are exposed to 27.6 % PM, and the remaining people are exposed to more than 72.4 %. The Gini Coefficient equals 0.56, showing inequity. Different from outdoor air quality studies (Jbaily et al., 2022; Rentschler and Leonova, 2023), passengers from cities with higher average incomes endure more severe PM exposure inside their subway systems. In contrast, the population in Shanghai, Tianjin, Mexico, and Beijing accounted for 48 % of the total population of the selected cities, and the cumulative exposure was only 14 %. A possible explanation is the newly built facilities and the PSDs, which effectively prevent the toxic $PM_{2.5}$ from dispersing. A sharp increase is observed at the point of Seoul. Cities such as Barcelona, Paris, Prague, and Toronto also had relatively high exposure. Zooming in the 50 % of the population with higher income, who bear 70 % of the total exposure, those passengers usually live in developed countries/regions (such as Spain, France, Canada, and the USA). In terms of the passenger load, these cities have a much lighter transit demand. While the high PM concentration led by the overused subway system, the aging of the system and lack of regular maintenance for both vehicles and the rail becomes the biggest contributor to the health burden of people inside the subway.

Special attention should be paid to subway workers, who are exposed to toxic pollution in the station on each working day. A study has shown that compared with the staff in the ticket hall, subway maintenance staff have the highest exposure to PM, followed by drivers and platform staff (Pan et al., 2019). Compared to subway passengers who temporarily stay in the subway system, subway workers suffer from much higher PM exposure with considerably higher health risks. These workers should, therefore, be given more protection strategies (e.g., masks and shorter work time), and subsidies to address adverse health effects potentially caused by occupational exposure.

6. Conclusion and perspectives

In the past two decades, subway IAQ has been widely investigated due to concerns about the impact of exposure to high PM concentrations on occupational and public health. Nevertheless, it continues to be an emerging topic of research where many important questions remain unanswered. Progress has been hampered in quantifying the adverse health effects of air pollutants in these environments due to transient public exposure patterns, diverse populations, and inconsistencies in different toxicological assays used. A deeper understanding of the factors that lead to elevated air pollutant concentrations, and consequently approaches to mediating these, is impeded by the complexity and lack of information concerning interacting components among outdoor air pollutants, subway emission sources, subway infrastructure, operational timetables, passenger density, and ventilation conditions.

Measurement studies have mostly focused on the mass concentration of $PM_{2.5}$ and PM_{10} in different subway systems. While some have quantified the chemical composition in time-integrated samples, few have bought more advanced high-time resolution particle size distribution and chemical composition measurement approaches that have been successfully used to better characterize the spatiotemporal variation in outdoor aerosol composition and could be used to improve our understanding of the relative importance of subway and outdoor source contributions, and other factors affecting concentrations in different networks. In addition, the ingress of outdoor air means that gaseous pollutants should not be ignored, and bioaerosols may be significant

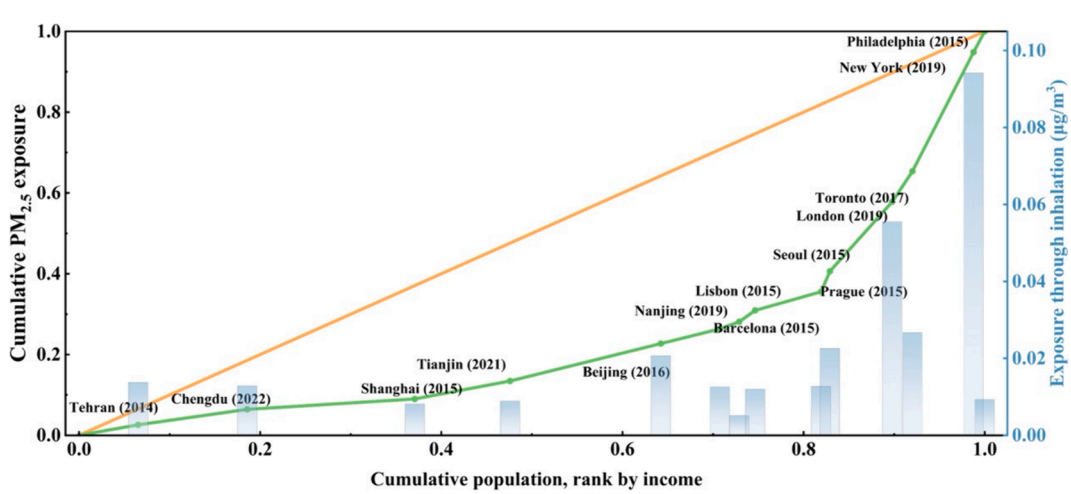

Fig. 7. Lorenz curve of the relationship between population and exposure for different cities worldwide. Exposure through inhalation was estimated based on the $PM_{2.5}$ concnetration in each city and the semi-empirical method (Eq. (2).

contributors in these enclosed environments, highlighting that there is an urgent need for further research.

Common best practices or standardization in measurement approaches should be developed to address the issue of measurement uncertainty, which helps the relative comparison among different measurement campaigns, as well as a precise capturing of the influencing factors on subway indoor air quality. Similar takeaways regarding the health impact assessment can also be concluded. The interpretation of toxicological evidence is hampered by an inconsistency in different toxicity assays employed and the mechanistic endpoints tested. The adoption of common approaches to assessing outdoor and subway PM in different countries would allow an interpretation of the relative toxicity of aerosol with different physicochemical properties found in subway microenvironments.

Finally, evidence should be gathered in different subway networks internationally to help justify future actions to reduce exposure and potential health risks of the large populations who use subways. Based on the collected data, modeling and simulations can be built to evaluate the effects of mitigation efforts on emissions and exposure inside the subway microenvironment, especially through high-cost practices such as infrastructure renovation, rolling stock renewal, the use of lower-emission materials, and improved HVAC systems.

CRediT authorship contribution statement

Shunyao Wang: Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization. **Tianchen Qin:** Writing – review & editing, Writing – original draft, Visualization, Investigation. **Ran Tu:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation. **Tianyuan Li:** Writing – review & editing, Conceptualization. **Gang I. Chen:** Writing – review & editing. **David C. Green:** Writing – review & editing, Conceptualization. **Xin Zhang:** Investigation. **Jialiang Feng:** Supervision, Resources. **Haobing Liu:** Writing – review & editing. **Ming Hu:** Visualization, Resources. **Qingyan Fu:** Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2024.108873.

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